FINESST Ancillary Documents

A. Facilities, Equipment, and Resources Statement

The proposed work requires modest, already-available resources; no facility development or shared-instrument time is needed. The proposing team has regular access to all resources below, so no resource-support letter is required.

Computing. A workstation with a single consumer GPU (NVIDIA GeForce GTX 1070 Ti, 8 GB) and 2 TB of local storage, owned by the research group and the FI, is sufficient for all model training and elevation processing in the proposal.

Workspace. The FI has office and lab space in NCALM at University of Houston with institutional backup storage for working data.

Software. The full processing stack is open source (Python scientific stack, PDAL, GDAL/OGR, WhiteboxTools, PyTorch, QGIS). No commercial licenses are required.

Field work. None proposed. The project is entirely archival-data-based; no Antarctic deployment, logistics support, or permitting is required.

B. Acknowledgements

The Future Investigator (FI) is the primary author of this proposal. The FI thanks Dr. Craig Glennie for scientific mentorship and guidance in shaping the research, and Cami Barlow, whose dissertation on Antarctic stream-channel detection and geomorphic change analysis provides the foundation this project builds upon. AI assistance (Anthropic's Claude, used as a drafting and editing tool under the FI's direction) contributed to text drafting, document formatting, and data-availability verification; all scientific content, claims, and decisions are the FI's own and were reviewed by the FI. Publicly archived datasets are credited to their providers: NASA ATM (via USGS), NCALM/OpenTopography, the Polar Geospatial Center (REMA), the McMurdo Dry Valleys LTER (via the Environmental Data Initiative), Copernicus/ECMWF (ERA5), and NCAR (AMPS).

C. Budget Justification / Narrative skeleton

Item	Amount	Justification
FI stipend		12-month graduate research assistantship at $2,601.67/month; the FI performs all proposed research
Tuition & fees		Consolidated tuition, 2 semesters × $3,852 (TX-resident Engineering-PhD rate, 9 SCH, AY2027); add mandatory fees
Conference travel		AGU (Fall Meeting), cryosphere section — poster in Yr 1, talk in later years
Publication		Open-access fee, one first-author paper
Computing/storage		Local storage expansion for full-valley raster stacks; no cloud compute required
Year 1 total (direct)		Under the $50,000/year cap

- No PI salary (FINESST does not fund the PI); no field-work or logistics costs.
- F&A/indirect: not permitted for scholarship awards; if the FI is treated as an employee (GRA), confirm allowable rates with the research office so the total stays under the $50,000/year cap.
- 3-year total stays under the $150,000 cap after stipend/tuition escalation.